

Dhrumil Rana

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EDUCATION

Master of Science in Data Science | @University of Texas at Arlington

January 2025 - December 2026

Bachelor of Engineering in Information Technology | @Government Engineering College Modasa

November 2020 - May 2024

EXPERIENCE

AI Engineer | NovachatAI

March 2026 – Present

- Redesigned **8+ Fanvue-focused dashboard modules** across onboarding, auth, settings, subscribers, scripts, messages, and Media Vault, improving responsiveness, workflow clarity, creator usability, and product polish.
- Built **media monetization workflows** including Media Vault labeling, gallery/list views, media previews, **PPV script media rendering**, subscriber filtering, and creator messaging UI for Fanvue-style sales flows.
- Resolved **15+ production-facing frontend issues**, including broken media previews, modal responsiveness, layout inconsistencies, dark/light theme bugs, onboarding progress, navigation polish, and mobile UI stability.
- Supported **post-breach security hardening** by defining frontend safeguards for auth validation, role-based access, RLS-aware UI behavior, exposed routes, and safer creator/admin workflows.

Software Developer Intern | Odoo

May 2024 – Sept 2024

- Developed **Python-based Odoo backend modules** for POS and inventory workflows, improving reliability, scalability, enterprise data consistency, and service-level maintainability.
- Optimized **PostgreSQL queries** and backend logic, achieving a **15% transaction speed increase** and resolving **50+ critical defects** in production environments.
- Designed **REST APIs** enabling real-time frontend-backend communication, improving responsiveness, system integration, and distributed business application workflows.

Research Analyst | Silent Info-tech Pvt. Ltd.

Jan 2024 – April 2024

- Built an end-to-end **Python ETL pipeline** processing **100k+ records** across **20+ features** for marketing analytics and data-driven decision-making.
- Integrated **OpenAI GPT APIs** to automate content generation workflows, reducing manual processing time by **80%** and improving team-level operational efficiency.
- Developed validation pipelines using **Pandas** and **NumPy**, improving data quality, consistency, reliability, and downstream analytics readiness.

PROJECTS

Aegis-Federate: Privacy-Preserving Multimodal Intelligence | GitHub

- Designed a cross-silo **federated learning system** using **Flower**, **gRPC**, and **Docker** for decentralized model training without sharing raw data.
- Built a **multimodal risk prediction architecture** combining **1D-CNN telemetry models** and **MLP-based EHR feature models**.
- Implemented **differential privacy with DP-SGD** using **Opacus** to reduce data leakage and model inversion risks.

Agentic-Cleanse Factory: Autonomous Data Quality Engine | GitHub

- Developed an **LLM-powered multi-agent ETL system** using **LangGraph** and **Groq** to automate data cleaning, validation, and transformation.
- Built autonomous agents for **outlier detection**, **schema mapping**, and heterogeneous dataset validation using prompt-engineering workflows.
- Reduced manual data processing effort by **80%**, improving pipeline efficiency and automated data correction reliability.

Twitter Scale Link Prediction Engine | GitHub

- Developed a graph embedding pipeline using **DeepWalk** with **1M+ random walks** on a **1.2M-node social graph**.
- Engineered a **1028-dimensional feature space** combining embeddings with **Adamic-Adar**, **Jaccard similarity**, and graph metrics.
- Trained a **Random Forest classifier** achieving **0.887 AUC**, outperforming traditional heuristic-based link prediction methods.

SKILLS

- Languages/Frontend/Backend:** Python, SQL, JavaScript, TypeScript, React, Next.js, Tailwind CSS, Supabase, PostgreSQL, REST APIs
- AI/ML/Data/Tools:** Machine Learning, Generative AI, RAG, LangChain, LangGraph, Groq, OpenAI APIs, Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch, Git, GitHub, Docker, Postman, Tableau, Power BI, Flower, gRPC, Opacus